MAJO BRIDGE TECHNOLOGIES AND EVENTS PLC

MaJo e-School Bridge (MayaBela) · INSA asset registration attachment

System & Architecture Design Document (SDD) — preferable format PDF

System and Architecture Design Document

Product: MaJo e-School Bridge (MayaBela) · asset MB-ESB-001 · version 1.0.6

Professionally sketched diagrams: Key components · System Architecture Diagram · Data Flow Diagram (DFD) · Data flow explanation

Document number	MB-SDD-001
Version	1.2 · 13 November 2026 · East Africa Time (Africa/Nairobi)
Replaces	1.1 of 13 November 2026 (PNG figures, but Level 1 had crossing arrows and A4 type was too small). 1.0 of 10 November 2026 was ASCII sketches rejected by Hibir.
INSA portal field	System & Architecture Design Document (SDD) — Key components, System Architecture Diagram, Data Flow Diagram (DFD), Data flow explanation. Recommendation received: add professionally sketched diagrams.
Companion	MB-SRS-001, MB-TDD-001, MB-UG-001, MB-SEC-001
Classification	CONFIDENTIAL (CMCSRS Annex B). No live secrets in this file.

Hibir completeness. This PDF contains the four required items by name: **Key components**, **System Architecture Diagram**, **Data Flow Diagram (DFD)**, and **Data flow explanation**, as professionally sketched figures (Figures 1–4) plus written explanation of each named flow.

1. Purpose

This document is the architecture baseline for INSA asset registration of MaJo e-School Bridge (MayaBela). It shows key components, how they interact, where data crosses a trust boundary, and which surfaces an attacker or a mis-scoped JWT can reach. Security requirements are in MB-SRS-001. This file is structure, diagrams, and flow.

The as-designed hosted architecture is described. Disk encryption keys on the current hosted database remain with the supplier (CMCSRS §6.29.f residual). A dedicated or on-prem instance for a school that cannot share this tenancy is the same application with a separate database URL — not a second product.

2. Key components

Key components of MaJo e-School Bridge are grouped in five layers. Presentation never stores student rows. Authorisation is the JWT plus row-level security (RLS), not the public anon key compiled into the client.

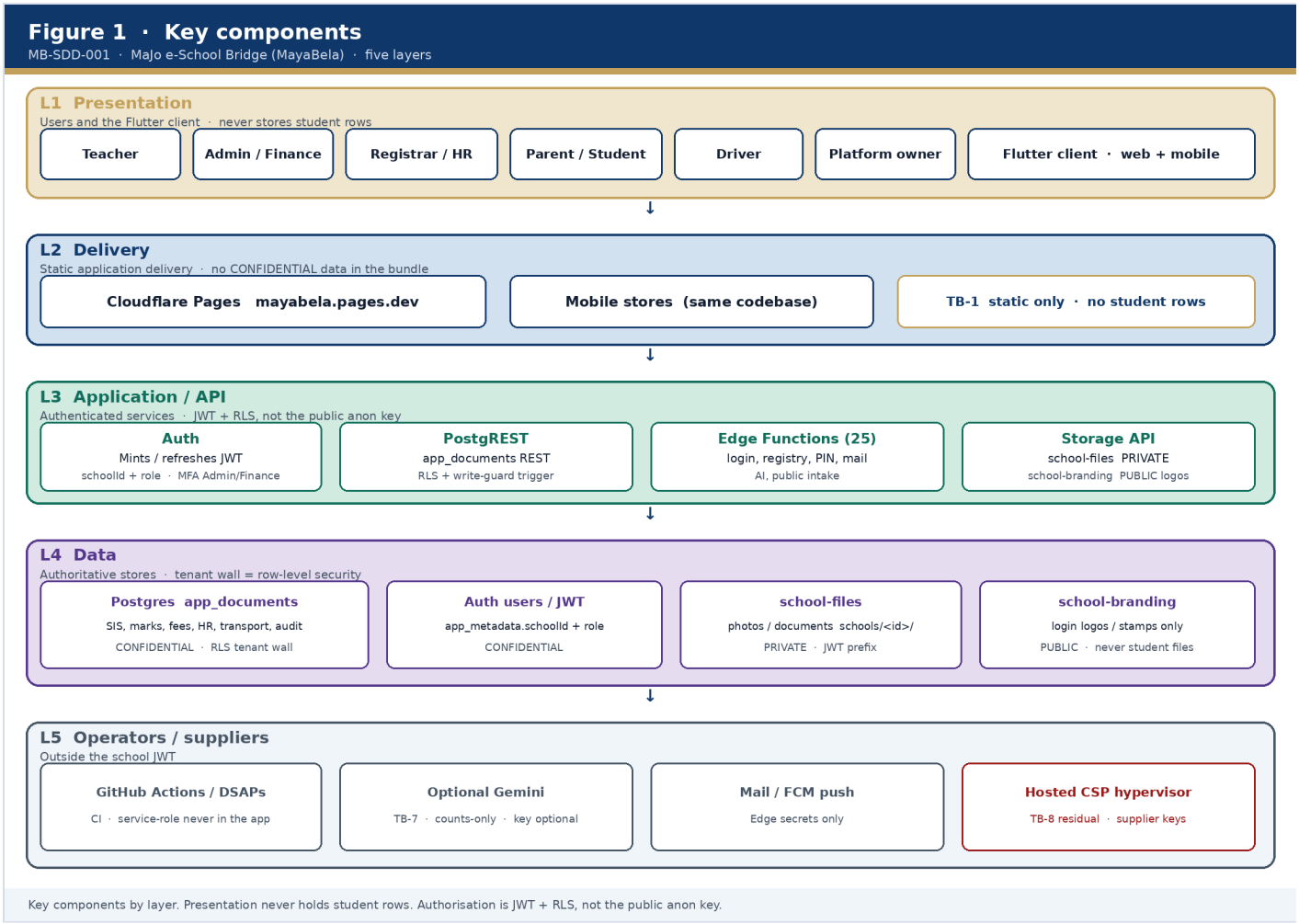


Figure 1 — Key components (professionally sketched). Layers L1–L5. Each layer is named on the figure.

Key component	Layer	What it does	Trust
Flutter client (web + mobile)	Presentation	One codebase for Teacher, Admin, Finance, Registrar, HR, Parent, Student, Driver, platform owner	Untrusted (TB-0)
Cloudflare Pages	Delivery	Static web at mayabela.pages.dev	TB-1 — no student rows in the JS bundle
Auth	Application	Mints JWT with schoolId + role in app_metadata. MFA for Admin and Finance	Inside TB-2
PostgREST	Application	REST on public.app_documents keyed by (collection, school_id, doc_id)	RLS + write-guard
Edge Functions (25)	Application	Login, registry, owner PIN, Maya Assistant, mail, public intake	Service-role only after a declared gate
Postgres app_documents	Data	SIS, marks, fees, HR, transport, audit	CONFIDENTIAL · tenant wall = RLS
school-files	Data	Photos and documents under schools/<id>/	Authenticated; prefix must match JWT school

Key component	Layer	What it does	Trust
school-branding	Data	Login logos and stamps	PUBLIC by design; no student files
Optional Gemini	Supplier	Maya Assistant, counts-only context	TB-7; key optional
Hosted CSP	Supplier	Hypervisor, disks, backups	TB-8 residual

3. System Architecture Diagram

The System Architecture Diagram shows users, the static delivery tier, the hosted backend, tenant isolation, and residual supplier trust. TLS is terminated by the hosted vendor. Mobile builds of the same Flutter codebase talk to the same TB-2 endpoints and do not embed a private CA.

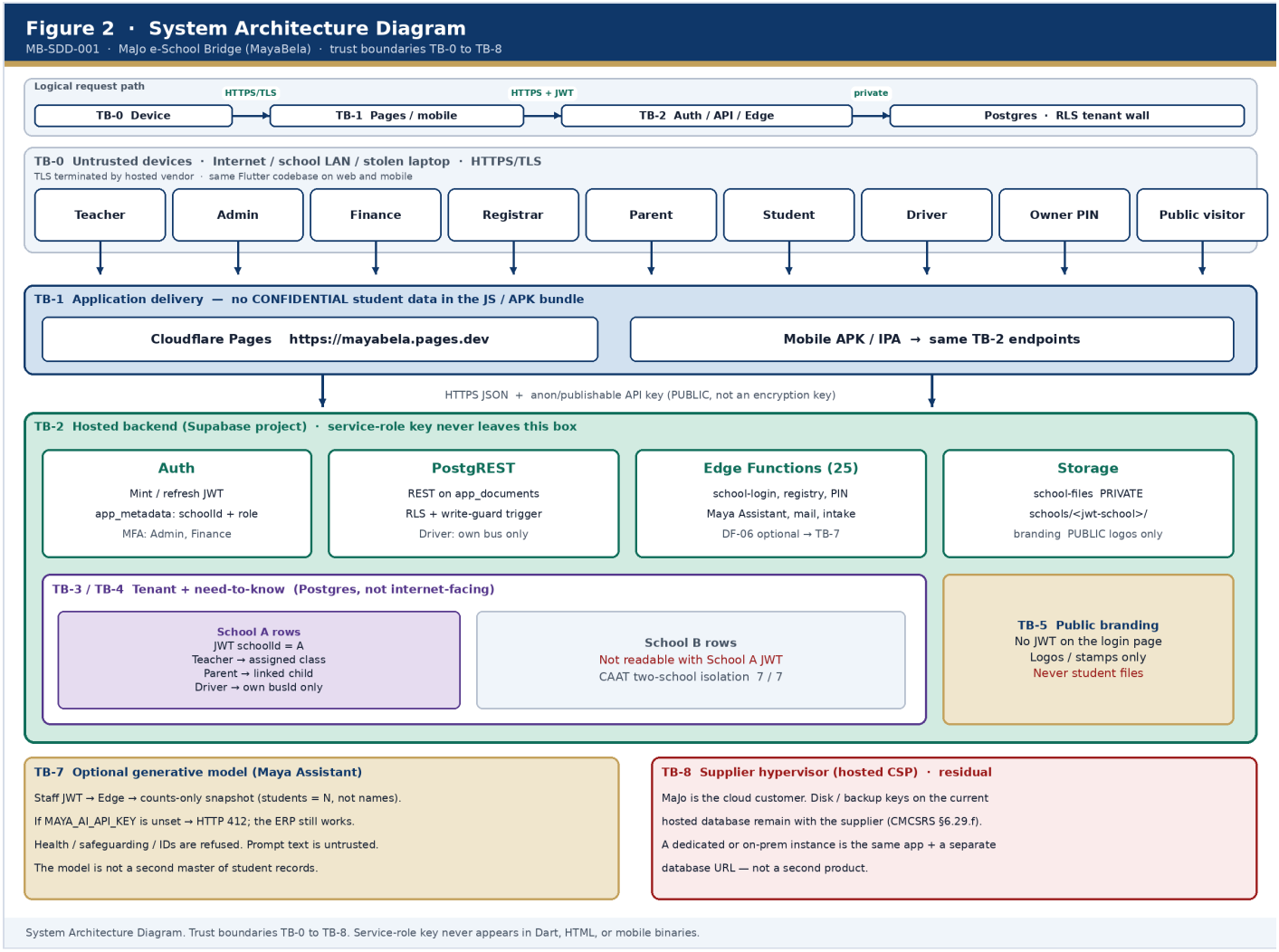


Figure 2 — System Architecture Diagram (professionally sketched). Trust boundaries TB-0 to TB-8.

3.1 Trust boundaries on the architecture

ID	Boundary	Untrusted side	Trusted side	What must hold
TB-0	Internet / user device	Browser, phone, stolen laptop	—	Treat as hostile. XSS, lost phone, shared teacher PC.
TB-1	Pages CDN	Anyone who can GET mayabela.pages.dev	Static assets + CSP/HSTS	No CONFIDENTIAL data in the static build except the public anon key.
TB-2	Hosted API	Any client with the anon key	Auth, PostgREST, Storage, Edge	No JWT → no app_documents. Service-role only inside Edge after a gate.
TB-3	School tenant	JWT of School A	Rows of School A	School A must not read School B (CAAT two-school 7/7 in source).
TB-4	Need-to-know	Teacher / driver / parent JWT	Class, bus, or linked child	Driver: own bus only. Teacher: assigned class; no payroll.
TB-5	Public storage	Unauthenticated GET	Logos only	Login page has no JWT. Student files must not live in branding.

ID	Boundary	Untrusted side	Trusted side	What must hold
TB-6	Platform owner	School JWT	Owner PIN + rate-limit	Provisioning must not use a teacher JWT. PIN is PBKDF2-SHA256.
TB-7	Optional AI	Prompt text	Gemini if key set	Counts-only; refuse health / safeguarding / IDs.
TB-8	Supplier hypervisor	MaJo as cloud customer	Host disks / backups	Supplier holds at-rest keys (§6.29.f). Not a Flutter control.

4. Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) is drawn in Yourdon / DeMarco notation: rectangles are external entities, circles are processes, open-ended bars are data stores, arrows are named data flows (DF-01 to DF-08). Figure 3 is the context (Level 0) diagram — external entities only, no data stores. Figure 4 decomposes process P0 into P1–P5 and shows stores D1–D4. Every arrow on the figures is the same identifier used in section 5.

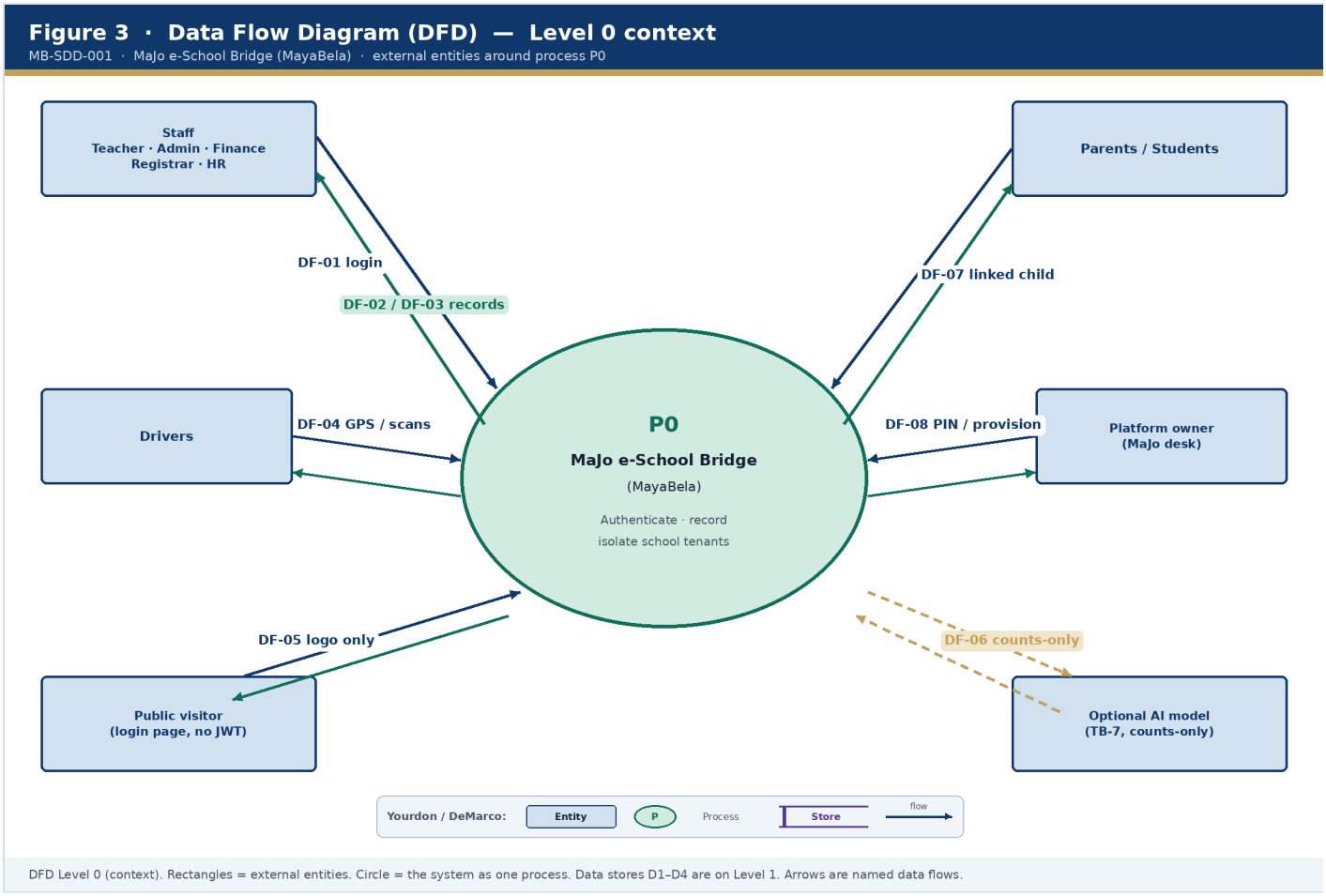
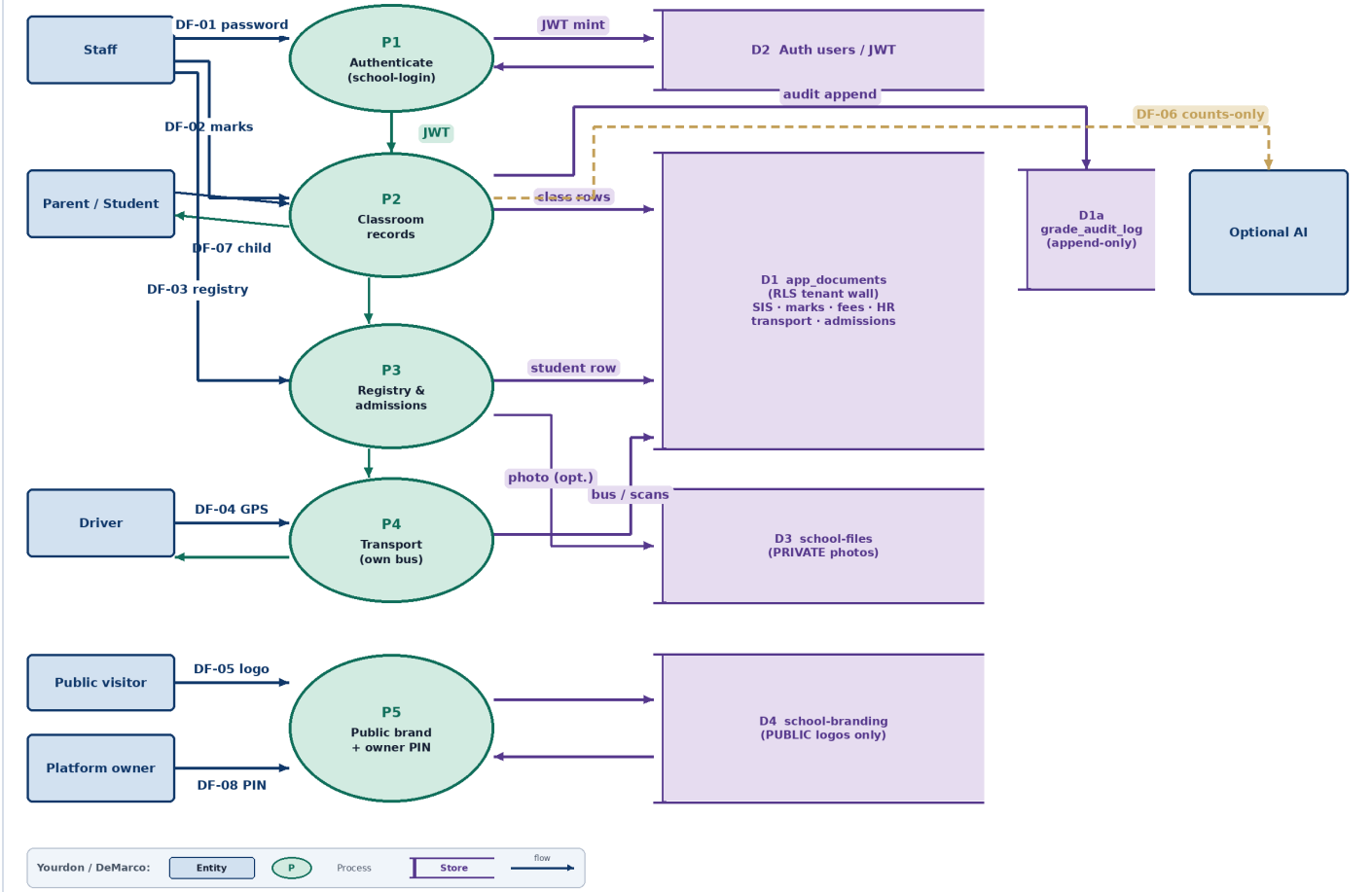


Figure 3 — Data Flow Diagram (DFD) Level 0 / context. External entities around process P0. Stores D1–D4 appear on Level 1.

Figure 4 · Data Flow Diagram (DFD) — Level 1
MB-SDD-001 · MaJo e-School Bridge (MayaBela) · P0 decomposed into P1–P5



DFD Level 1. Processes P1–P5, stores D1–D4. No store-to-store flows. Registry student rows go to D1 via P3; classroom writes go to D1 via P2 + RLS.

Figure 4 — Data Flow Diagram (DFD) Level 1. Processes P1–P5, stores D1–D4. No store-to-store flows. DF-01 to DF-08 match section 5.

4.1 DFD dictionary

ID	Type	Name	Notes
P0	Process	MaJo e-School Bridge	Context process
P1	Process	Authenticate	school-login; bcrypt; JWT mint; MFA for Admin/Finance
P2	Process	Classroom records	PostgREST INSERT/UPDATE; RLS + write-guard; grade_audit_log
P3	Process	Registry and admissions	Edge school-upsert-registry; desk gate; service-role after gate
P4	Process	Transport	Driver JWT; own busId only
P5	Process	Public brand + owner PIN	No student lists; PIN is not a school JWT
D1	Data store	app_documents	CONFIDENTIAL SIS / marks / fees / HR / transport / audit
D2	Data store	Auth users / JWT	schoolId + role in app_metadata
D3	Data store	school-files	PRIVATE under schools/<id>/
D4	Data store	school-branding	PUBLIC logos only

5. Data flow explanation

This Data flow explanation names every principal flow on the Data Flow Diagram (DFD). Each flow states source, destination, data, protocol, and the control that keeps children's records inside the correct school and role.

5.1 DF-01 School login (password) — P1

A device on TB-0 sends HTTPS to Edge function `school-login` (public, rate-limited). The function verifies the bcrypt secret for schoolId + username + role. Auth then mints a JWT with `app_metadata.schoolId` and `role` (store D2). The device stores the JWT and uses it on later calls to PostgREST, Storage, and other Edge functions (TB-2, TB-3, TB-4). Privileged MFA (Admin, Finance): after password success, Settings requires TOTP enrolment. Teachers are not blocked from class if MFA is unset.

5.2 DF-02 Ordinary classroom write — P2

A teacher JWT is sent over HTTPS to PostgREST for INSERT/UPDATE on `app_documents` (store D1). RLS requires `school_id = jwt_school_id()`. The write-guard trigger further checks collection and assigned class. Marks also append to `grade_audit_log`. A modified client still hits the database wall (TB-3), not only the Flutter UI.

5.3 DF-03 Privileged registry write — P3

Staff JWT is sent over HTTPS to `school-upsert-registry`. The function verifies the JWT, then `guardRegistryRecord` (registrar desk for admissions). Only after that gate does it use the service-role to upsert D1. The service-role bypasses RLS, so the application gate replaces RLS for this path. CI fails if an Edge folder is added without a declared auth mode.

5.4 DF-04 Driver location and scans — P4

A driver JWT writes `bus_live_positions` and `transport_scans` in D1. RLS plus a driver trigger require `transportId == busId`. Other buses and walker students are not readable. Assigned-bus passengers and own GPS remain available.

5.5 DF-05 Public branding (no JWT) — P5

Anyone may GET `school-public-brand` (rate-limited) and/or the public storage object under D4. The response is school name plus logo URL. No emails, passwords, or student lists. The login page cannot present a JWT, so logos are public by design. Student files stay in D3, never D4.

5.6 DF-06 Maya Assistant (optional) — TB-7

A staff JWT calls `maya-assistant-chat`. The function builds a counts-only snapshot (for example students = N, not names). If `MAYA_AI_API_KEY` is set, the prompt crosses TB-7 to the model. Safeguarding, health, password, and ID prompts are refused. If the key is unset the function returns 412 and the ERP still works.

5.7 DF-07 Parent / student read

A parent JWT may read only linked-child rows in D1. A student JWT reads own portal rows. Neither role writes payroll, HR, or other-school records.

5.8 DF-08 Platform owner provisioning — P5 / TB-6

The MaJo desk uses an owner PIN (PBKDF2-SHA256, rate-limited), not a school JWT, to list/create/update schools and upload logos. Provisioning must not be possible with a teacher token.

5.9 What never flows

- The service-role key never flows to Dart, HTML, or mobile binaries.

- School A JWT never flows to School B rows (TB-3).
- Named student, health, counselling, safeguarding, draft marks, and unpublished report cards are not stored as a cloud master copy outside the School Node / hosted Postgres of that tenant.
- Maya Assistant does not receive a dump of children’s identities.

6. Logical structure (narrative of the architecture)

6.1 Presentation

One Flutter application (web on Cloudflare Pages; mobile stores). The anon / publishable API key is compiled into the client (PUBLIC). It is not an encryption key. Authorisation is the JWT plus RLS. The service-role key must never appear in Dart, HTML, or mobile binaries.

6.2 Application / API

- **PostgREST** on `public.app_documents` keyed by `(collection, school_id, doc_id)`. RLS and a write-guard trigger enforce school and role. Driver SELECT is further restricted to assigned-bus students.
- **Edge Functions** (25) when the client cannot use RLS alone (password verify, PIN, service-role writes, public intake). Each function has a declared auth mode in `edge_auth_inventory.ts`; CI fails if a folder is added without one.

6.3 Data

Store	Content	Annex B	Trust
Postgres app_documents	SIS, marks, fees, HR, transport, audit	CONFIDENTIAL	Inside TB-2; RLS is the tenant wall
Auth users / JWT	schoolId + role in app_metadata	CONFIDENTIAL	DB helpers ignore user-editable metadata
school-files	Photos, documents under schools/<id>/	CONFIDENTIAL	Authenticated; prefix must match JWT school
school-branding	Login logos / stamps	PUBLIC	Anon SELECT; no student files
Edge secrets	Service-role, mail, optional AI key, push secret	CONFIDENTIAL	TB-2 only; DSAPs

7. Attack surfaces

Surfaces are listed so INSA can see what we expose, not as a pentest letter (that letter is still scheduled after staging apply).

ID	Surface	Who can reach it	Primary controls	Residual
AS-01	Static web (Pages)	Internet	HTTPS, HSTS, CSP, nosniff, DENY framing	XSS in third-party map/font origins allowed by CSP
AS-02	Anon API key in the client	Anyone who downloads the app	RLS; anon has no app_documents policy	Inherent to hosted anon model
AS-03	school-login / password reset / register-parent / submit-application	Internet, no JWT	Rate-limit; hashed secrets; no user-existence leak on reset	Credential stuffing if school passwords are weak
AS-04	PostgREST with a stolen staff JWT	Anyone with that token until expiry	TB-3/TB-4 RLS; MFA on Admin/Finance	Teacher JWT theft on a shared PC — MFA not forced on teachers
AS-05	Storage public branding	Internet	Logos only by policy	Mis-upload of a student file into branding would be a data spill (IRP)
AS-06	Storage school-files	Authenticated JWT	Prefix schools/<jwt school>/	Object ACL bugs would be High

ID	Surface	Who can reach it	Primary controls	Residual
AS-07	Edge with service-role	After JWT/PIN/public gate	Inventory + registry policy + PIN PBKDF2	A missed gate is equivalent to RLS bypass — CI tracks inventory
AS-08	Platform owner PIN	Internet to those functions	Rate-limit; PBKDF2; no school JWT	PIN reuse / shoulder-surfing at the vendor desk
AS-09	send-push	Caller with x-push-secret	Secret; re-reads notification row	Secret in Edge config
AS-10	Maya Assistant	Staff JWT	Counts-only; refuse sensitive; optional key	User may still type a name into the prompt — instruction refuses to echo PII
AS-11	GitHub Actions / DSAPs	MaJo operators	Secrets in CI; production never the RLS test target	Human error; DSAP clearance files still national-scheme residual
AS-12	Supplier platform	Hosted CSP	DPA; no destructive tests on live project	Foreign host; keys not MaJo-held

8. Edge Function auth modes

Mode	Functions	Trust assumption
password-login	school-login	Secret verified; then JWT
user-jwt	refresh-claims, change-password, upsert-account, upsert-registry, delete-account, verify-student, maya-assistant-chat	Caller JWT; school isolation; extra desk checks where noted
platform-owner-pin	owner-pin, list/create/update/delete-school, upload-logo, upload-mark, mail-config	PIN, not a school JWT
public-rate-limited	password-reset pair, student-reset request, register-parent, invite-parent, submit-application, public-brand, client-crash-report	No JWT; rate-limit; no CONFIDENTIAL dump in the response
push-secret	send-push	Shared Edge secret + stored notification row

9. Security design decisions

- **RLS in the database, not only the UI.** A modified client still hits TB-3.
- **Anon key public; service-role private.** Losing the anon key is not a breach if RLS holds.
- **Login logos public.** The login page cannot present a JWT; student files stay in a private bucket.
- **Teachers not forced onto MFA.** Privileged desks (Admin, Finance) are. Classroom availability is a stated requirement (MB-SRS-001).
- **One operator, many schools.** Isolation is JWT + RLS, not per-tenant disk keys handed to schools.

10. Related documents

MB-SRS-001 (requirements). MB-TDD-001. MB-UG-001. MB-SEC-001. CSMS-AST-001 (assets). CSMS-KEY-001 (crypto). CSMS-SUP-001 (cloud register). CSMS-DPIA-001 (AI). CSMS-IRP-001 (incidents). CAAT: `supabase/tests/local_pg`.

Prepared by (Engineering / CSO)

Majo Bridge Technologies and Events PLC · 13 November 2026, East Africa Time

For INSA asset registration — Hibir field SDD

Re-upload this PDF: INSA_SDD_MaJo_eSchool_Bridge.pdf

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Contains Key components, System Architecture Diagram, Data Flow Diagram (DFD), and Data flow explanation.
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